

SARAL SINGHAL

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Education

Vellore Institute of Technology

2023 – 2027

B.Tech., Computer Science and Engineering, CGPA: 8.42/10

Technical Skills

Languages and Database: Python, Java, SQL, Supabase

Data Analytics: Pandas, NumPy, Matplotlib, Power BI, Excel

ML Libraries: Scikit-Learn, XGBoost, TensorFlow, PyTorch, Keras, Streamlit

Concepts: ETL, Data Modelling, Requirement Gathering, Stakeholder Management, KPI Management

Tools: Git, GitHub, Jupyter Notebook, VS Code, Power Automate, n8n

Experience

Deloitte Touche Tohmatsu India LLP

Gurugram, India

Intern, Strategy, Risk & Transactions

May 2026 – August 2026

- **Feasibility Study:** Contributed to the feasibility study for a 10 MW Tier III data centre in India under a public-private partnership (PPP) model, covering demand assessment, site and infrastructure requirements and a 20-year operating case.
- **Bid Compliance Analytics:** Designed a Power BI dashboard on a star-schema data model with custom DAX measures to score 19 bidders against a 32-point NHAI tender checklist, replacing a manual document-by-document review process.
- **Asset Monetisation Reporting:** Developed the reporting layer for the Asset Registration System, consolidating dispersed highway asset records into review-ready dashboards for stakeholder sign-off.

Ernst & Young LLP

Delhi, India

Research Intern

May 2025 – June 2025

- **Revenue Analysis:** Quantified non-fare box revenue streams for Chennai Metro Rail Limited (CMRL), benchmarking commercial space utilisation and land value capture across stations.
- **Requirements Design:** Specified functional requirements and workflow models for a platform automating vendor-based non-fare revenue collection, tracking and reconciliation.
- **Geospatial Data:** Mapped Chhattisgarh and Uttar Pradesh national highway stretches through GIS, standardising asset records for downstream registration and analysis.

Projects

BleedDetectX

[Link](#)

Python, PyTorch, TensorFlow, Keras

- **Model Development:** Built and compared U-Net, ResNet-UNet and DMP-WCEBleedNet architectures for simultaneous bleeding classification and segmentation in Wireless Capsule Endoscopy imagery.
- **Results:** Reached a Dice coefficient of 0.812 and 98.31% classification accuracy with DMP-WCEBleedNet, outperforming the U-Net baseline on the held-out test set.
- **Evaluation:** Assessed models across Dice, IoU, Precision, Recall and F1 on a class-imbalanced dataset.

TeleTrace

[Link](#)

Python, SQLite, SQL

- **Pipeline Design:** Built a five-stage ETL pipeline in pure Python and SQLite by generating, validating, cleaning along with loading and analysing the processing 500 synthetic call-centre records seeded with missing fields, duplicates and malformed timestamps, where every rejected record retained a failure reason alongside a run-level ingestion audit log.
- **Modelling & Analysis:** Normalised timestamps to IST and derived duration, hour and weekend attributes with imputation explicitly flagged; loaded into an idempotent SQLite schema so that re-runs never duplicated rows, and answered business questions in SQL using conditional aggregation with CSV report exports.

Certifications & Achievements

- **Microsoft Certified: Fabric Data Engineer Associate** — data engineering with Spark, pipelines and Lakehouse.
- **University of Michigan: Applied Machine Learning in Python.**
- Published and presented a research paper on Wireless Capsule Endoscopy at ICNACT 2026.

Extracurricular Activities

Co-Lead, Vitronix (Technical Club)

- Scripted technical workshops and speaker sessions for a 300+ member community, coordinating design, outreach and logistics teams to ship content on schedule and support 110+ registrations.

Co-Lead, Mahro Rajasthan Club (Cultural Club)

- Led a 7-member team across 4 cultural events attended by 500+ students; standardised a bilingual script template and two-round review process that cut last-minute rewrites.